Ahana Kaur

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Professional Summary

Machine Learning Engineer candidate with strong hands-on experience in NLP, Computer Vision, and Speech Processing. Proven record of building end-to-end ML systems including data pipelines, training, evaluation, and deployment using PyTorch, TensorFlow, and FastAPI. Experienced with MLOps workflows, CI/CD automation, and real-time inference systems.

Education

Bennett University — Greater Noida, India
Master of Computer Applications (MCA) — Currently 4th Semester — *2024–2026*

- Relevant Coursework: Statistical Machine Learning, Deep Learning, NLP, Soft Computing, Image & Video Processing

Presidency University — Bangalore, India
Bachelor of Computer Applications (BCA) — *2021–2024*

- Relevant Coursework: Machine Learning, Data Structures & Algorithms, Programming in Python/C

Technical Skills & Soft Skills

Programming: Python
Machine Learning: PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, OpenCV, NLTK
Deep Learning: CNNs, Transformers, Encoder–Decoder, Sequence-to-Sequence Models
Speech & Vision: Whisper, Pyannote, ESPnet, MediaPipe, Torchaudio
MLOps & Deployment: Docker, Jenkins, FastAPI, Flask, pytest, CI/CD, Git
Data Processing: Feature Engineering, Data Augmentation, Tokenization, Evaluation Metrics
Soft Skills: Analytical problem solving, self-directed learning, systems thinking, technical communication, research-oriented mindset, ownership and execution

Selected ML Projects

Offline AI-Powered Meeting Transcription & Summarization System (Refining) | *Whisper, Pyannote, Transformers, PyTorch*

- Built an end-to-end speech analytics pipeline supporting multilingual ASR with **90%+ transcription accuracy** on clean audio samples.
- Implemented speaker diarization using Pyannote, achieving **accurate segmentation for up to 6 speakers** with timestamp alignment.
- Reduced inference latency by **35%** through audio chunking, silence removal, and denoising preprocessing.
- Generated abstractive summaries using Transformer models, condensing **30–60 min meetings into 5–7 key bullet insights**.
- Designed the system to run **fully offline on local hardware**, eliminating cloud dependency and reducing operational costs to **zero**.

NoiseFloor — Intelligent Job Discovery System | *Python, NLP, Web Scraping, SQLite, Twilio, Flask, Cloud*

- Designed and built an intelligent job discovery system that filters noise from job platforms, reducing irrelevant listings by **70–80%** using relevance scoring and legitimacy analysis.
- Implemented **semantic resume–JD matching** and rule-based risk detection to identify scams, vague postings, and suspicious application flows, providing **explainable signals** instead of binary filtering.
- Developed **persistent deduplication using SQLite**, preventing repeated job alerts across runs and prioritizing newly posted roles, eliminating daily duplicate notifications entirely.
- Integrated **real-time WhatsApp alerts via Twilio**, including interactive **Like / Dislike feedback buttons**, enabling human-in-the-loop preference capture for future personalization.
- Deployed the system on the cloud (Render) using **event-driven webhooks and scheduled scraping**, enabling **24/7 autonomous operation** without manual intervention.

AI Tutor / Educational Assistant | *Python, BART, Hugging Face, Flask, SQLite*

- Built document ingestion pipelines processing **PDFs, PPTs, and images** with automated text extraction and cleanup.
- Implemented BART-based abstractive summarization, reducing raw content length by **60–70%** while preserving key concepts.
- Generated personalized revision notes and quizzes, improving content readability and structured learning flow.

Real-Time Emotion Recognition System | *TensorFlow, OpenCV, CNNs*

- Trained CNN-based emotion classifiers on facial image datasets, achieving **70% validation accuracy** across 7 emotion classes.
- Implemented real-time webcam inference with **15–20 FPS** performance on consumer hardware.
- Logged emotion predictions at **5-second intervals** for temporal behavior analysis and pattern tracking.

Neural Machine Translation (English–Hindi) | *MarianMT, Hugging Face, PyTorch*

- Implemented Transformer-based encoder–decoder translation using MarianMT for English–Hindi language pairs.
- Achieved fluent sentence-level translations with **low-latency inference** suitable for interactive use cases.
- Built an interactive interface to visualize source text, translated output, and model responses in real time.

50 Days, 50 Machine Learning Projects (Ongoing)

- Applied ML challenge covering regression, classification, clustering, evaluation, and experimentation workflows.
- Maintained structured documentation and experiments for reproducibility and learning tracking.

Personal AI Assistant (Ongoing)

- Developing a local AI voice assistant for system control, task automation, and device-level interactions.

End-to-End MLOps Pipeline for ML API Deployment | *Jenkins, Docker, FastAPI, Scikit-learn*

- Designed CI/CD pipelines using Jenkins to automate **model testing, validation, and deployment** across environments.
- Containerized ML inference APIs with Docker, ensuring **100% reproducible builds** and consistent runtime behavior.
- Deployed FastAPI-based prediction services capable of handling **concurrent real-time requests** with structured JSON responses.

Hobbies

- Building personal AI & automation projects
- Self-driven coding challenges (50 Days, 50 Projects — multiple tracks)
- Digital & traditional drawing, cooking, reading mythology and history

Experience

Tata Group — Remote

Data Analytics Job Simulation (Forage) — *Dec 2025*

- Performed GenAI-assisted exploratory data analysis to assess data quality and identify customer risk indicators.
- Designed a no-code predictive framework for delinquency risk assessment and an AI-driven collections strategy aligned with ethical AI and regulatory compliance.

Certifications

- Generative AI: Introduction and Applications – IBM
- Programming for Everybody (Python) – University of Michigan
- Computer Networking – Google
- Artificial Intelligence for Cyber Security – Huang Quy La (Ongoing)
- Python Mega Course: Build 20 Apps – Ardit Sulce (Ongoing)